

BEE 4750 Homework 5: Mixed Integer and Stochastic Programming

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Due Date

Thursday, 12/04/24, 9:00pm

Overview

Instructions

- In Problem 1, you will use mixed integer programming to solve a waste load allocation problem.
- In Problem 2, you will formulate a stochastic optimization problem.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
Pkg.add("CSV")
```

```
Activating project at `~/hw5-ero23_hw5`
Resolving package versions...
No Changes to `~/hw5-ero23_hw5/Project.toml`
No Changes to `~/hw5-ero23_hw5/Manifest.toml`
```

```

using JuMP
using HiGHS
using DataFrames
using GraphRecipes
using Plots
using Measures
using MarkdownTables
using CSV

```

Problems (Total: 30 Points)

Problem 1 (24 points)

Three cities are developing a coordinated municipal solid waste (MSW) disposal plan. Three disposal alternatives are being considered: a landfill (LF), a materials recycling facility (MRF), and a waste-to-energy facility (WTE). The capacities of these facilities and the fees for operation and disposal are provided below.

- **LF:** Capacity 200 Mg, fixed cost \$2000/day, tipping cost \$50/Mg;
- **MRF:** Capacity 350 Mg, fixed cost \$1500/day, tipping cost \$7/Mg, recycling cost \$40/Mg recycled;
- **WTE:** Capacity 210 Mg, fixed cost \$2500/day, tipping cost \$60/Mg;

The MRF recycling rate is 40%, and the ash fraction of non-recycled waste is 16% and of recycled waste is 14%. Transportation costs are \$1.5/Mg-km, and the relative distances between the cities and facilities are provided in the table below.

City/Facility	Landfill (km)	MRF (km)	WTE (km)
1	5	30	15
2	15	25	10
3	13	45	20
LF	-	32	18
MRF	32	-	15
WTE	18	15	-

The fixed costs associated with the disposal options are incurred only if the particular disposal option is implemented. The three cities produce 100, 90, and 120 Mg/day of solid waste, respectively, with the composition provided in the table below.

Component	% of total mass	Combustion ash (%)	MRF Recycling rate (%)
Food Wastes	15	8	0
Paper & Cardboard	40	7	55
Plastics	5	5	15
Textiles	3	10	10
Rubber, Leather	2	15	0
Wood	5	2	30
Yard Wastes	18	2	40
Glass	4	100	60
Ferrous	2	100	75
Aluminum	2	100	80
Other Metal	1	100	50
Miscellaneous	3	70	0

The information in the above table will help you determine the overall recycling and ash fractions. Note that the recycling residuals, which may be sent to either landfill or the WTE, have different ash content than the ash content of the original MSW. You will need to determine these fractions to construct your mass balance constraints.

Reminder: Use `round(x; digits=n)` to report values to the appropriate precision!

Problem 1.1

Based on the information above, calculate the overall recycling and ash fractions for the waste produced by each city.

```
RECYCLE_rate= (.15*0+.4*.55+.05*.15+.03*.1+.02*0+.05*.3+.18*.4+.04*.6+.02*.75+.02*.8+.01*.5+
println("Recycling fraction = ", round(RECYCLE_rate; digits=4), '%')
ASH_rate = (.15*.08+.4*.07+.05*.05+.03*.1+.02*.15+.05*.02+.18*.02+.04*1+.02*1+.02*1+.01*1+.03*
println("Ash fraction = ", round(ASH_rate; digits=4), '%')
```

Recycling fraction = 37.75%

Ash fraction = 16.41%

Problem 1.2

What are the decision variables for your optimization problem? Provide notation and variable meaning.

I am using similar decision variables to what we used in the class example...

W_{ij} = Waste transported from city i to disposal j (Mg/day)

$cities = (1,2,3)$
 $disposal = (MRF, WTE, LF)$
 $R(1,2,3) = \text{Residual waste transported (MRF-WTE , MRF-LF , WTE-LF) (Mg/day)}$
 $Y_j = \text{Operational status (on/off) of disposal } j \text{ (binary)}$

Problem 1.3

Formulate the objective function. Make sure to include any needed derivations or justifications for your equation(s).

```
Full_MRF_Tipping = 7+(.3775*40) #accounting for only 37.75% of the Mg being recyclable
println("MRF Tipping Cost= ", round(Full_MRF_Tipping; digits=4), '%')
```

MRF Tipping Cost= 22.1%

broadly the objective is:
 minimize (W_{ij}, Y)

$\text{sum}(\text{fixed_cost}[j] * Y[j] \text{ for } j \text{ in facilities}) + \text{sum}(\text{tipping}[j] * W[i,j] \text{ for } i \text{ in cities, } j \text{ in facilities})$
 $+ \text{transport} * \text{sum}(\text{dist}[(i,j)] * W[i,j] \text{ for } i \text{ in cities, } j \text{ in facilities})$

Because I updated the MRF tipping cost above I don't need to add anything special about recycling and can just use that value in the tipping cost calculations!

I also included my calculations and the exact objective below in the image

$W_{city, plant}^{(1,2,3)}, (MRF, WTE, LF) \quad R^{(1-3)} (MRF \rightarrow WTE, MRF \rightarrow LF, WTE \rightarrow LF)$

MRF cost $\rightarrow 1500 \cdot y_1 + 22.1 (W_{11}^{MRF} + W_{21}^{MRF} + W_{31}^{MRF})$
 (1)

WTE cost $\rightarrow 2500 \cdot y_2 + 60 (W_{12}^{WTE} + W_{22}^{WTE} + W_{32}^{WTE} + R_{MRF, WTE}^{1,2})$
 (2)

LF cost $\rightarrow 2000 \cdot y_3 + 50 (W_{13}^{LF} + W_{23}^{LF} + W_{33}^{LF} + R_{MRF \rightarrow LF}^{1,3} + R_{WTE \rightarrow LF}^{2,3})$
 (3)

transport $\rightarrow 1.5 (W_{11} \cdot 30 + W_{12} \cdot 15 + W_{13} \cdot 5 + W_{21} \cdot 25 + W_{22} \cdot 10 + W_{23} \cdot 15 + W_{31} \cdot 45 + W_{32} \cdot 20 + W_{33} \cdot 13 + R_1 \cdot 15 + R_2 \cdot 32 + R_3 \cdot 18)$

$\rightarrow$
 min $67.1 W_{11} + 59.6 W_{21} + 89.6 W_{31} + 82.5 W_{12} + 75 W_{22} + 90 W_{32} + 57.5 W_{13} + 72.5 W_{23} + 69.5 W_{33} + 82.5 R_1 + 98 R_2 + 77 R_3 + 1500 y_1 + 2500 y_2 + 2000 y_3$
 w, R, y

constraints: $W_{11} + W_{12} + W_{13} = 100$
 (waste amount...)
 $W_{21} + W_{22} + W_{23} = 90$
 $W_{31} + W_{32} + W_{33} = 120$
 $(WTE \rightarrow LF)$
 $R_3 = .1641 (W_{12} + W_{22} + W_{32} + R_1)$
 $R_1 + R_2 = .6225 (W_{11} + W_{21} + W_{31})$

MRF: $W_{11} + W_{21} + W_{31} \leq 350$
 WTE: $W_{12} + W_{22} + W_{32} + R_1 \leq 210$
 LF: $W_{13} + W_{23} + W_{33} + R_2 + R_3 \leq 200$

$y_1 = \begin{cases} 0 & \text{if } W_{11} + W_{21} + W_{31} = 0 \\ 1 & \end{cases}$
 $y_2 = \begin{cases} 0 & \text{if } W_{12} + W_{22} + W_{32} + R_1 = 0 \\ 1 & \end{cases}$
 $y_3 = 1$ (because some waste is sent to landfill no matter what)

$W_{ij}, R_1, R_2, R_3 \geq 0$

Problem 1.4

Derive all relevant constraints. Make sure to include any needed justifications or derivations.

The constraints are in the work shown above!

Problem 1.5

Find the optimal solution (using JuMP to solve the problem). Report the optimal objective value.

```
waste_model = Model(HiGHS.Optimizer) # initialize model object

#-----set up...
cities = 1:3
plants = 1:3 # 1=MRF, 2=WTE, 3=LF

#-----non-negativity constraints
@variable(waste_model, W[cities, plants] >= 0) # waste flows
@variable(waste_model, R[1:3] >= 0) # residuals:
#R[1]=MRF->WTE , R[2]=MRF->LF , R[3]=WTE->LF

# Syntax from the slides in class!
@variable(waste_model, Y[plants], Bin)

#Waste totals!
@constraint(waste_model, W[1,1] + W[1,2] + W[1,3] == 100)
@constraint(waste_model, W[2,1] + W[2,2] + W[2,3] == 90)
@constraint(waste_model, W[3,1] + W[3,2] + W[3,3] == 120)

#Residuals from WTE
@constraint(waste_model,
    R[3] == 0.1641 * (W[1,2] + W[2,2] + W[3,2] + R[1]) #0.1641 = ash rate
)
#Residuals from the MRF
@constraint(waste_model,
    R[1] + R[2] == 0.6225 * (W[1,1] + W[2,1] + W[3,1]) #0.6225 = 1-recycle rate
)
#Plant/Disposal Capacities
@constraint(waste_model,
    (W[1,1] + W[2,1] + W[3,1]) <= 350 * Y[1]
)
@constraint(waste_model,
```

```

        (W[1,2] + W[2,2] + W[3,2] + R[1]) <= 210 * Y[2]
    )
    @constraint(waste_model,
        (W[1,3] + W[2,3] + W[3,3] + R[2] + R[3]) <= 200 * Y[3]
    )

    # MRF open condition
    @constraint(waste_model, sum(W[:,1]) <= 1000 * Y[1])

    # WTE open condition
    @constraint(waste_model, sum(W[:,2]) + R[1] <= 1000 * Y[2])

    # LF must always be open
    @constraint(waste_model, Y[3] == 1)

    #My simplified objective function!
    @objective(waste_model, Min,
        67.1*W[1,1] + 59.6*W[2,1] + 89.6*W[3,1] +
        82.5*W[1,2] + 75.0*W[2,2] + 90.0*W[3,2] +
        57.5*W[1,3] + 72.5*W[2,3] + 69.5*W[3,3] +
        82.5*R[1] + 98.0*R[2] + 77.0*R[3] +
        1500*Y[1] + 2500*Y[2] + 2000*Y[3]
    )
    print(waste_model)

```

Min 67.1 W[1,1] + 59.6 W[2,1] + 89.6 W[3,1] + 82.5 W[1,2] + 75 W[2,2] + 90 W[3,2] + 57.5 W[1,3] + 72.5 W[2,3] + 69.5 W[3,3] + 82.5 R[1] + 98.0 R[2] + 77.0 R[3] + 1500 Y[1] + 2500 Y[2] + 2000 Y[3]

Subject to

$$W[1,1] + W[1,2] + W[1,3] = 100$$

$$W[2,1] + W[2,2] + W[2,3] = 90$$

$$W[3,1] + W[3,2] + W[3,3] = 120$$

$$-0.1641 W[1,2] - 0.1641 W[2,2] - 0.1641 W[3,2] - 0.1641 R[1] + R[3] = 0$$

$$-0.6225 W[1,1] - 0.6225 W[2,1] - 0.6225 W[3,1] + R[1] + R[2] = 0$$

$$Y[3] = 1$$

$$W[1,1] + W[2,1] + W[3,1] - 350 Y[1] = 0$$

$$W[1,2] + W[2,2] + W[3,2] + R[1] - 210 Y[2] = 0$$

$$W[1,3] + W[2,3] + W[3,3] + R[2] + R[3] - 200 Y[3] = 0$$

$$W[1,1] + W[2,1] + W[3,1] - 1000 Y[1] = 0$$

$$W[1,2] + W[2,2] + W[3,2] + R[1] - 1000 Y[2] = 0$$

$$W[1,1] = 0$$

$$W[2,1] = 0$$

$$W[3,1] = 0$$

$$W[1,2] = 0$$

```

W[2,2]  0
W[3,2]  0
W[1,3]  0
W[2,3]  0
W[3,3]  0
R[1]    0
R[2]    0
R[3]    0
Y[1]    binary
Y[2]    binary
Y[3]    binary

```

```
optimize!(waste_model)
```

Running HiGHS 1.12.0 (git hash: 755a8e027): Copyright (c) 2025 HiGHS under MIT licence terms
MIP has 11 rows; 15 cols; 44 nonzeros; 3 integer variables (3 binary)

Coefficient ranges:

```

Matrix  [2e-01, 1e+03]
Cost     [6e+01, 2e+03]
Bound    [1e+00, 1e+00]
RHS      [1e+00, 1e+02]

```

Presolving model

10 rows, 14 cols, 42 nonzeros 0s

8 rows, 13 cols, 36 nonzeros 0s

Presolve reductions: rows 8(-3); columns 13(-2); nonzeros 36(-8)

Solving MIP model with:

8 rows

13 cols (2 binary, 0 integer, 0 implied int., 11 continuous, 0 domain fixed)

36 nonzeros

Src: B => Branching; C => Central rounding; F => Feasibility pump; H => Heuristic;

I => Shifting; J => Feasibility jump; L => Sub-MIP; P => Empty MIP; R => Randomized rounding;

S => Solve LP; T => Evaluate node; U => Unbounded; X => User solution; Y => HiGHS solution;

Z => ZI Round; l => Trivial lower; p => Trivial point; u => Trivial upper; z => Trivial upper bound

Nodes			B&B Tree		Objective Bounds			Dynamic Cuts	
Src	Proc.	InQueue	Leaves	Expl.	BestBound	BestSol	Gap	Cuts	Int.
J	0	0	0	0.00%	-inf	31740.783	Large	0	
S	0	0	0	0.00%	2000	27855.482115	92.82%	0	

50.0% inactive integer columns, restarting

Model after restart has 1 row, 3 cols (0 bin., 0 int., 0 impl., 3 cont., 0 dom.fix.), and 3 r

	0	0	0	0.00%	26922.085548	27855.482115	3.35%	0
T	0	0	0	0.00%	26922.085548	27855.482115	3.35%	0
	1	0	1	100.00%	27855.482115	27855.482115	0.00%	0

Solving report

Status	Optimal
Primal bound	27855.4821151
Dual bound	27855.4821151
Gap	0% (tolerance: 0.01%)
P-D integral	0.00110394708635
Solution status	feasible
	27855.4821151 (objective)
	0 (bound viol.)
	0 (int. viol.)
	0 (row viol.)
Timing	0.01
Max sub-MIP depth	0
Nodes	1
Repair LPs	0
LP iterations	6
	0 (strong br.)
	0 (separation)
	0 (heuristics)

```
# Print the optimal objective value
println("Optimal objective: ", round(objective_value(waste_model), digits =2))

# Print waste flows
for i in 1:3, j in 1:3
    println("W[$i,$j] = ", round(value(W[i,j]); digits=2))
end

# Print residuals
for k in 1:3
    println("R[$k] = ", round(value(R[k]),digits =2))
end

# Print plant open/close decisions
for j in 1:3
```

```
println("Y[$j] = ", round(value(Y[j]), digits =2))
end
```

Optimal objective: 27855.48

```
W[1,1] = 0.0
W[1,2] = 0.0
W[1,3] = 100.0
W[2,1] = 0.0
W[2,2] = 90.0
W[2,3] = 0.0
W[3,1] = 0.0
W[3,2] = 41.6
W[3,3] = 78.4
R[1] = 0.0
R[2] = 0.0
R[3] = 21.6
Y[1] = 0.0
Y[2] = 1.0
Y[3] = 1.0
```

Problem 1.6

Draw a diagram showing the flows of waste between the cities and the facilities. Which facilities (if any) will not be used? Does this solution make sense?

```
#CHAT GPT used to format the graph
using GraphPlot, Colors

# ----- Nodes -----
names = ["City1", "City2", "City3", "MRF", "WTE", "LF"]
shapes = [:hexagon, :hexagon, :hexagon, :rect, :rect, :rect]
xpos = [-2, 0, 2, -1.5, 0.5, 0]
ypos = [2, 2, 2, 0, 0, -1.5]

# ----- Adjacency matrix -----
A = [
    0 0 0 1 1 1;
    0 0 0 1 1 1;
    0 0 0 1 1 1;
    0 0 0 0 1 1;
    0 0 0 0 0 1;
```

```

    0 0 0 0 0 0
]

# ----- Edge labels -----
edge_labels = Dict(
    (1,4) => "", (1,5) => "", (1,6) => "100",
    (2,4) => "", (2,5) => "90", (2,6) => "",
    (3,4) => "", (3,5) => "41.6", (3,6) => "78.4",
    (4,5) => "", (4,6) => "", (5,6) => "21.6"
)

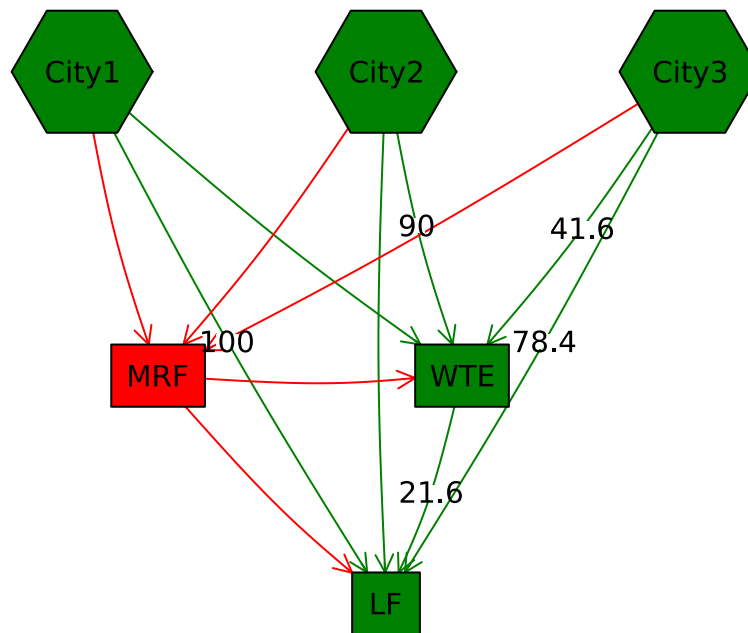
# ----- Edge colors (red if going to or coming from MRF, green if flow, gray if 0) -----
edge_colors = Array{Colorant}(undef, size(A)...)
for i in 1:size(A,1)
    for j in 1:size(A,2)
        if i == 4 || j == 4
            # Edge connected to MRF
            edge_colors[i,j] = colorant"red"
        elseif A[i,j] != 0
            # Flow exists but not connected to MRF
            edge_colors[i,j] = colorant"green"
        else
            # No flow
            edge_colors[i,j] = colorant"gray"
        end
    end
end

# ----- Node colors (MRF red, others green) -----
node_colors = [colorant"green", colorant"green", colorant"green",
               colorant"red", colorant"green", colorant"green"]

# ----- Plot -----
p = graphplot(A, names=names, edgelabel=edge_labels,
              markersize=0.4, markershapes=shapes,
              markercolor=node_colors, x=xpos, y=ypos,
              fontsize=10, edgecolor=edge_colors)

display(p)

```



This didn't really make sense to me at first. The MRF plant wasn't used at all despite having the lowest costs for every aspect. Having thought about it more I think this could be due to distances, or the high residual rate out of the MRF, because if less than half of the material could actually be recycled then it was being sent there just to be transported to another disposal location after, meaning adding even more costs. It was still very surprising for me to see that result.

Problem 2 (6 points)

Consider a two-period economic dispatch problem, based on the multi-period example from Lecture 14 (on 10/29). The generator data, including ramping constraints for each generator, is provided in 'data/generators.csv.' In period 1, the demand is $d_1 = 1100\text{MW}$. In period 2, the demand is projected to be $d_2 = 1200\text{MW}$, but there is a 25% probability that it is \$1500. In the first period, the solar capacity factor is 0.9 and the wind capacity factor is 0.45, but in the second period, there is some uncertainty: the forecasted solar and wind capacity factors are 0.95 and 0.4, respectively, but there is a 30% probability that they are 0.75 and 0.5. Your goal is to identify how to dispatch your generators to minimize the cost of meeting demand.

Problem 2.1

Draw a scenario tree for this problem.

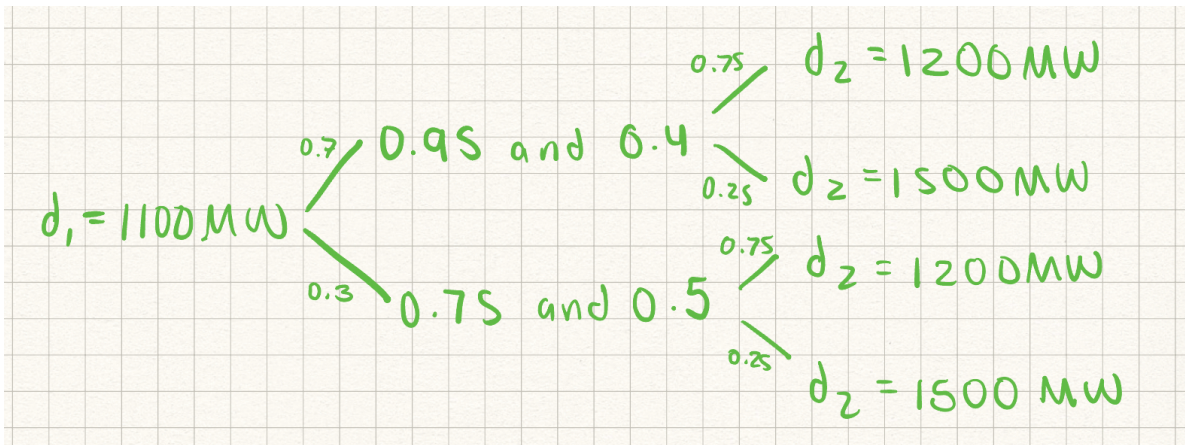


Figure 1: Scenario Tree!

Problem 2.2

Formulate a stochastic linear program for this problem based on your scenario tree from Problem 2.1 and the data in `data/generators.csv`.

```
gens = DataFrame(CSV.File("data/generators.csv"))

plant = gens.Plant
cost = Dict(g => gens.VarCost[gens.Plant == g][1] for g in plant)
Pmin = Dict(g => gens.Pmin[gens.Plant == g][1] for g in plant)
Pmax = Dict(g => gens.Pmax[gens.Plant == g][1] for g in plant)
Ramp = Dict(g => gens.Ramp[gens.Plant == g][1] for g in plant)
res = Dict(g => gens.Resource[gens.Plant == g][1] for g in plant)

#period 1 vals
d1 = 1100.0
cf_s1 = 0.90
cf_w1 = 0.45

#period 2 vals
scenarios = [
    ("s1", 1200, 0.95, 0.40, 0.75*0.70), #corresponding prob = .525
    ("s2", 1200, 0.75, 0.50, 0.75*0.30), # .225
    ("s3", 1500, 0.95, 0.40, 0.25*0.70), # .175
```

```

    ("s4", 1500, 0.75, 0.50, 0.25*0.30), # .075
]
d2_exp    = sum(s[2]*s[5] for s in scenarios)
cf_s2_exp = sum(s[3]*s[5] for s in scenarios)
cf_w2_exp = sum(s[4]*s[5] for s in scenarios)

model = Model(HiGHS.Optimizer)
@variable(model, g1[g in plant] >= 0)
@variable(model, g2[g in plant] >= 0)

@objective(model, Min,
    sum(cost[g]*g1[g] for g in plant) +
    sum(cost[g]*g2[g] for g in plant)
)

#period 1
for g in plant
    if res[g] == "Wind"
        @constraint(model, g1[g] <= Pmax[g]*cf_w1)
    elseif res[g] == "Solar"
        @constraint(model, g1[g] <= Pmax[g]*cf_s1)
    else
        @constraint(model, Pmin[g] <= g1[g] <= Pmax[g])
    end
end
#period 2
for g in plant
    if res[g] == "Wind"
        @constraint(model, g2[g] <= Pmax[g]*cf_w2_exp)
    elseif res[g] == "Solar"
        @constraint(model, g2[g] <= Pmax[g]*cf_s2_exp)
    else
        @constraint(model, Pmin[g] <= g2[g] <= Pmax[g])
    end
end

for g in plant
    @constraint(model, g2[g] - g1[g] <= Ramp[g])
    @constraint(model, g1[g] - g2[g] <= Ramp[g])
end

# Power balance

```

```

@constraint(model, sum(g1[g] for g in plant) == d1)
@constraint(model, sum(g2[g] for g in plant) == d2_exp)

optimize!(model)

```

Running HiGHS 1.12.0 (git hash: 755a8e027): Copyright (c) 2025 HiGHS under MIT licence terms

LP has 30 rows; 14 cols; 56 nonzeros

Coefficient ranges:

Matrix [1e+00, 1e+00]

Cost [5e+00, 4e+02]

Bound [0e+00, 0e+00]

RHS [1e+02, 1e+03]

Presolving model

12 rows, 12 cols, 28 nonzeros 0s

3 rows, 8 cols, 10 nonzeros 0s

1 rows, 2 cols, 2 nonzeros 0s

Presolve reductions: rows 1(-29); columns 2(-12); nonzeros 2(-54)

Solving the presolved LP

Performed postsolve

Solving the original LP from the solution after postsolve

```

Model status      : Optimal
Objective value    : 1.7720000000e+04
P-D objective error : 0.0000000000e+00
HiGHS run time     : 0.01

```

Warning: thread = 1 warning: parsed expected 6 columns, but didn't reach end of line around

@ CSV /Users/elianaolson/.julia/packages/CSV/XLcqT/src/file.jl:593

Warning: thread = 1 warning: only found 6 / 7 columns around data row: 4. Filling remaining

@ CSV /Users/elianaolson/.julia/packages/CSV/XLcqT/src/file.jl:592

Warning: thread = 1 warning: only found 6 / 7 columns around data row: 5. Filling remaining

@ CSV /Users/elianaolson/.julia/packages/CSV/XLcqT/src/file.jl:592

Warning: thread = 1 warning: only found 6 / 7 columns around data row: 6. Filling remaining

@ CSV /Users/elianaolson/.julia/packages/CSV/XLcqT/src/file.jl:592

```

println("\n--- Period 1 Dispatch ---")
for g in plant
    println(g, ": ", round(value(g1[g]), digits =4))
end

```

```
println("\n--- Period 2 Dispatch ---")
for g in plant
    println(g, ": ", round(value(g2[g]), digits =4))
end

println("\n--- Expected Total Cost ---")
println(objective_value(model))
```

--- Period 1 Dispatch ---

Biomass: 0.0

Hydroelectric: 195.0

Geothermal: 0.0

NG CCGT: 220.0

NG CT: 100.0

Wind: 135.0

Solar: 450.0

--- Period 2 Dispatch ---

Biomass: 0.0

Hydroelectric: 381.0

Geothermal: 0.0

NG CCGT: 220.0

NG CT: 100.0

Wind: 129.0

Solar: 445.0

--- Expected Total Cost ---

17720.0

References

List any external references consulted, including classmates.

I used ChatGPT to help format and figure out syntax for figures and outputs in cleaner ways.